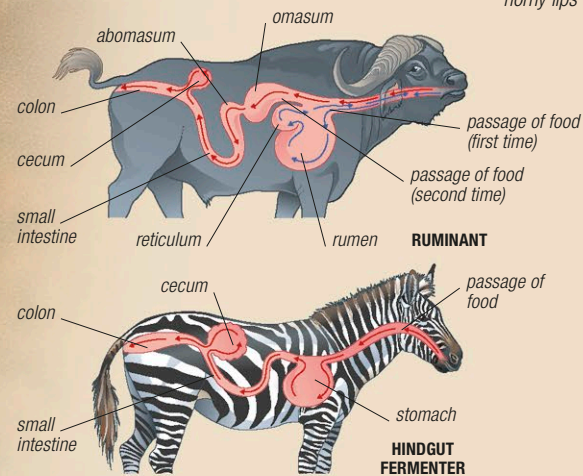


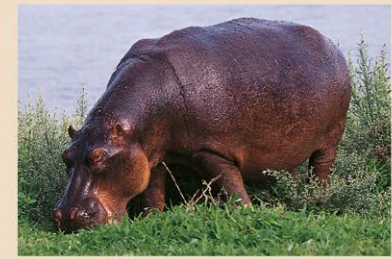
Feeding

Almost all hoofed mammals are herbivores. The plants they eat contain indigestible cellulose (the major component of plant cell walls), which is split into digestible carbohydrates via bacterial fermentation. In ruminants (see below), food passes slowly through their system to maximize the nutrition gained. These animals thrive where food is limited but of high quality. In hindgut fermenters (see below), food is not retained in the stomach and passes through the system more quickly. These animals live where food is plentiful but of poor quality. As a result, a larger volume of food must be eaten to obtain enough nutrients.



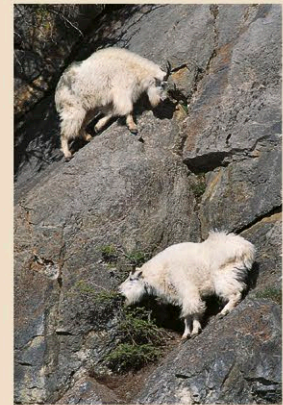
DIGESTIVE SYSTEMS

Ruminants, such as buffaloes, have a complex stomach. In the first chamber, the rumen, bacterial fermentation occurs; food is then regurgitated, re-chewed ("chewing the cud"), and swallowed, this time passing through the digestive system. Hindgut fermenters, such as zebras, have a simple stomach: the fermenting bacteria are in the cecum and at the start of the colon.



GRAZING

Grazers feed almost exclusively on grass. To find fresh pasture, some hoofed mammals migrate long distances. The hippopotamus (above) usually feeds at night, using its horny lips to crop the grass.



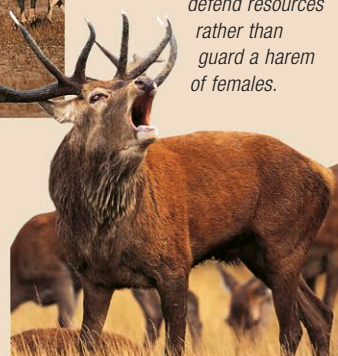
BROWSING

Browsers eat almost any plant material. These mountain goats spend much of their time in mountainous areas where grass is scarce—they supplement their diet by eating mosses, lichens, herbs, and woody plants.



DEFENDING A HAREM

During the breeding season, some male hoofed mammals, such as this red deer, gather a harem, which they defend fiercely. The loud roaring and large antlers of the stag help deter rivals.



DEFENDING RESOURCES

Hoofed mammals living in arid areas, such as these onagers, often form mixed-sex herds. Males tend to defend resources rather than guard a harem of females.

Social systems

Hoofed mammals have a variety of social systems, depending on factors such as habitat, body size, whether breeding is seasonal, and whether they migrate. The following examples are typical hoofed mammal social organizations. Tapirs, rhinoceroses, and some forest antelopes are mostly solitary, and a male's territory covers that of several females. Dik-diks live in pairs that occupy small territories. In gazelles, males set up small territories, and female herds wander in and out of these areas. Hartebeests and most zebras live in harems consisting of a male and a number of females. Red deer, on the other hand, form separate-sex herds except during the breeding season (which is known as the "rut"). Males fight with each other, and the most successful will collect a large harem of females.



5

NUTRITIOUS MEAL

The mother eats the afterbirth. Once this is done, she will lick the calf to stimulate it.



6

PROTECTIVE INSTINCTS

A newborn calf is vulnerable, and the mother is reluctant to let even another wildebeest approach.



7

STANDING UP

The calf attempts to stand. In less than 45 minutes, it will be able to run, reducing the risk of predation.



8

REJOINING THE HERD

Mother and calf quickly rejoin the herd: in a group, the young calf is less conspicuous.